Spectral Analysis

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Introduction

Spectral analysis examines a time series in the frequency domain rather than the time domain. Periodicities appear as peaks in the power spectrum, slow trends dominate the low-frequency end, and white noise has a flat spectrum. The view is complementary to autocorrelation-based time-domain analysis: the spectral density and the autocovariance function are Fourier-transform pairs, so neither view contains information the other lacks, but each makes different features easier to see.

Prerequisites

A working understanding of Fourier analysis, the autocovariance function, and the time-frequency duality of stationary processes.

Theory

For a stationary process with autocovariance function γ_k, the spectral density is

$$S(f) = \sum_{k=-\infty}^{\infty} \gamma_k e^{-2\pi i f k},$$

defined on $f \in [-1/2, 1/2]$ for unit-spaced data. The periodogram is the sample-based estimator, computed as the squared magnitude of the FFT of the centred series divided by the series length. Raw periodograms are inconsistent (variance does not decrease with n), so smoothing — Bartlett, Welch, kernel-based, or multi-taper — is applied to produce stable spectral estimates.

A peak at frequency f indicates a cyclic component with period $1/f$ time units; the area under a frequency band is the variance contribution from that band.

Assumptions

The series is approximately stationary (or has been made so by detrending and differencing); sufficient length for frequency resolution $1/n$ to distinguish nearby peaks.

R Implementation

```r
# Simulated series with two cycles
set.seed(2026)
t <- 1:500
y <- sin(2 * pi * t / 12) + 0.5 * sin(2 * pi * t / 4) + rnorm(500, 0, 0.3)

# Periodogram
spec_p <- spectrum(y, plot = FALSE)
plot(spec_p, main = "Raw periodogram")

# Smoothed
```

```r
spec_s <- spectrum(y, spans = c(5, 5), plot = FALSE)
plot(spec_s, main = "Smoothed periodogram")
abline(v = 1/12, col = "red", lty = 2)
abline(v = 1/4,  col = "red", lty = 2)
```

Output & Results

`spectrum()` returns the periodogram with frequencies and spectral-density estimates, plus the chosen smoothing-window structure. The plot shows distinct peaks at $f = 1/12$ and $f = 1/4$, matching the simulated cycles.

Interpretation

A reporting sentence: "Spectral analysis of the simulated series with two embedded cycles produced peaks at frequencies $f = 0.083$ (period 12) and $f = 0.25$ (period 4) on the smoothed periodogram with spans = c(5, 5); the relative peak heights match the relative amplitudes of the simulated components." Always state the smoothing window in the figure caption; raw periodograms are noisy and often misread.

Practical Tips

- Smoothing (`spans = c(5, 5)` or similar) trades variance for bias; tune to balance peak detection against resolution.
- Frequency resolution is $1/n$; for closely spaced periodicities, longer series are essential.
- Use a log-log plot of frequency vs spectral density to expose power-law behaviour and broadband features.
- Multi-taper estimation (`multitaper::spec.mtm`) provides better bias-variance trade-off than spans-based smoothing for short series.
- Spectral analysis complements ACF / PACF; use both, especially when looking for cyclic versus trend structure.
- For non-stationary series, time-frequency methods (short-time Fourier, wavelets) generalise the stationary spectral framework.

R Packages Used

Base R `spectrum()` for periodograms with smoothing; `multitaper::spec.mtm()` for multi-taper estimation; `astsa::mvspec()` for multivariate spectral analysis; `WaveletComp` for time-frequency localisation when stationarity fails.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Time series basics